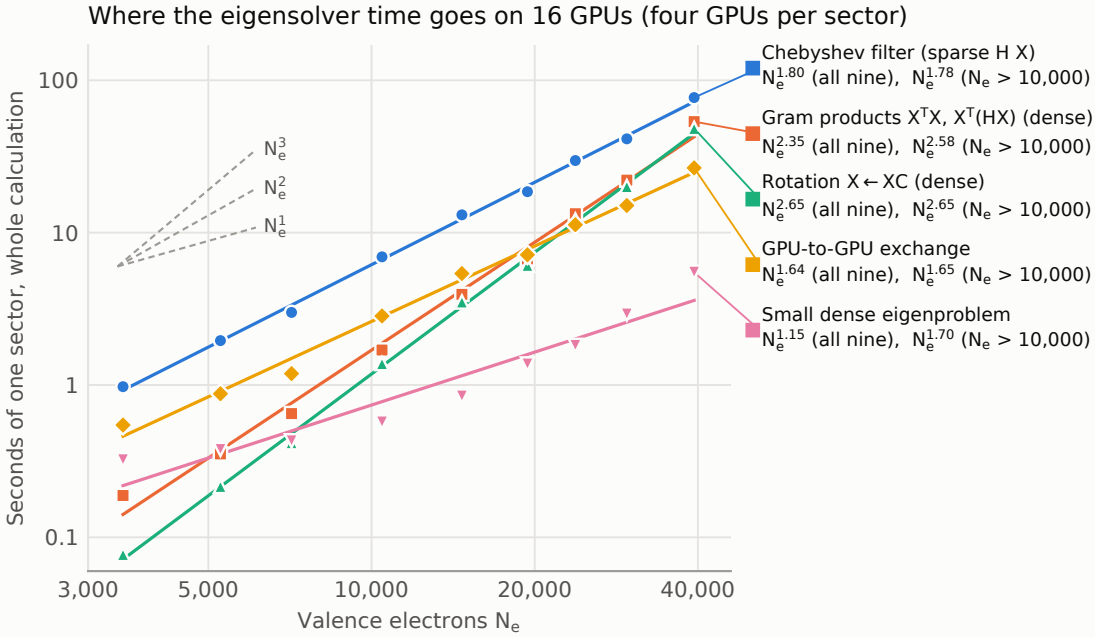
Where the eigensolver time goes on 16 GPUs (four GPUs per sector)
Seconds of one sector, whole calculation
100
10
1
0.1
3,000
5,000
10,000
20,000
40,000
Valence electrons N_e
N_e^3
N_e^2
N_e^1
Chebyshev filter (sparse H X)
$N_e^{1.80}$ (all nine), $N_e^{1.78}$ (N_e > 10,000)
Gram products X^TX, $X^T(HX)$ (dense)
$N_e^{2.35}$ (all nine), $N_e^{2.58}$ (N_e > 10,000)
Rotation $X \leftarrow XC$ (dense)
$N_e^{2.65}$ (all nine), $N_e^{2.65}$ (N_e > 10,000)
GPU-to-GPU exchange
$N_e^{1.64}$ (all nine), $N_e^{1.65}$ (N_e > 10,000)
Small dense eigenproblem
$N_e^{1.15}$ (all nine), $N_e^{1.70}$ (N_e > 10,000)